

COMBINATORICS ASSIGNMENT 2

ROB TOMPKINS

6. Prove that

$$P(n, k) = k! \binom{n}{k}$$

Solution.

□

10. Prove that:

$$\sum_{k=0}^n r^{n-k} \binom{n}{k} = (r+1)^n$$

Solution.

□

16. Prove Vandermonde's Identity:

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

Solution.

□

19. Prove the identity:

$$\binom{n+1}{r+1} = \sum_{k=r}^n \binom{k}{r}$$

Solution.

□

20. Prove the identity:

$$2_n - 1 = \sum_{k=0}^{n-1} 2^k$$

Solution.

□

22. Prove the identity:

$$\binom{n}{k} \binom{k}{t} = \binom{n}{t} \binom{n-t}{k-t}$$

Solution.

□

24. Prove the identity:

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r+k} = \binom{m+n}{m+r}$$

Solution.

□

26. Prove the identity:

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}$$

Solution.

□

33. Prove the identity:

$$\sum_{k=n-s}^{m-r} \binom{m-k}{r} \binom{n+k}{s} = \binom{m+n+1}{r+s+1}$$

Solution.

□